

Disproof of Conjecture 00000003836

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Pinned convention

The classical signature algorithm encodes a reading word by signs and repeatedly deletes adjacent $+-$ pairs. Equivalently, scan left to right with a stack: a minus sign cancels a plus sign at the top of the stack; otherwise the new sign is retained. The conjecture says that the number of remaining plus signs is invariant under every cyclic rotation.

Two-letter counterexample

Take

$$w = (+, -), \quad \text{rot}(w) = (-, +).$$

For w , the unique adjacent pair is $+-$ and cancels. Hence the reduced word is empty and its plus count is 0. For the rotation, no $+-$ pair occurs, so the reduced word is $(-, +)$ and its plus count is 1.

Proposition 1. *The plus count of the signature output is not invariant under cyclic rotation.*

Proof. The two explicit reductions above give $0 \neq 1$. □

This refutes the first conjunct, and therefore the filed conjunction, without needing any interpretation of the separately mentioned string parameter. If an author uses the opposite cancellation convention ($-+$ instead of $+-$), the very same two cyclic rotations exchange their roles and still give counts 0 and 1.

The accompanying Lean file implements the stack algorithm and kernel-checks all of these finite evaluations.